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Calcium Chloride

Mechanism of action

- Calcium is the active ingredient in calcium chloride.
- Calcium is a mineral that is essential for a number of normal body functions including cell membrane function, enzyme reactions, transmission of nerve impulses, cardiac electrophysiology, contraction of cardiac and skeletal muscle, and coagulation.
- Calcium raises the cardiac action potential threshold and protects cardiac cell membranes from the effects of hyperkalaemia, resulting in a reduction in dysrhythmias associated with hyperkalaemia.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Release syndrome following crush injury in adults.
- ✓ Suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to suspected hyperkalaemia.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but calcium should be administered if indicated.

Dosage

- Crush injury:

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- 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
- Repeat the dose if signs of hyperkalaemia persist or recur.
- Seek clinical advice for a child.

- Hyperkalaemia with severe ECG changes:
  - 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
  - Repeat the dose after 20 minutes if signs of severe ECG changes persist or recur.
  - Seek clinical advice for a child.

- Cardiac arrest secondary to hyperkalaemia:
  - 6.8 mmol (1 g) of calcium chloride IV as a bolus for an adult.
  - Repeat the dose after 10 minutes if ROSC is not achieved.
  - Seek clinical advice for a child.

Administration

- Crush injury and hyperkalaemia with severe ECG changes:
  - Administer IV over 1 minute into a large vein via a running line, as this reduces venous irritation.

- Cardiac arrest secondary to hyperkalaemia:
  - Administer as a bolus via a running line.

- Do not mix with other medicines (and in particular do not mix with sodium bicarbonate) as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum flush of 50 ml of 0.9% sodium chloride between medicines.

Common adverse effects

- Venous irritation including redness and pain at the site of injection.
- Tingling sensation.
- Rapid administration may cause dysrhythmias.

Usual onset of effect

- 2-5 minutes.

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Usual duration of effect

- 1-4 hours.

Usual preparation

- Ampoule containing 6.8 mmol (1 g) in 10 ml.

Pharmacokinetics

- 80% is excreted in faeces and 20% is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Document the dose administered in mmol.
- Calcium gluconate is an alternative formulation of calcium, usually containing 2.2 mmol/10 ml. Administer two ampoules at a time if only calcium gluconate is available.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated. Information under dosage and administration headings updated.